



Flat Knitting

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Background

1.1 Context

This document describes the modelling approach for the **flat knitting process** for a functional unit of **1 kg of knitted fabric ("coarse knitted tricot")**.

It is designed to understand:

- which data sources were used,
- how the three scenarios (best / average / worst) were constructed,
- how yarn losses, lubricants, electricity, waste and capital goods were modelled,
- how indirect energy (HVAC, compressors, lighting, cleaning systems) was handled.

1.2 Scope

- **Functional unit:** 1 kg of knitted fabric.
- **Technology:** flat knitting (mix of typical machine types; mid-level scenario).
- **Included:**
 - lubricants and associated emissions,
 - electricity consumption (direct and indirect from the process),
 - capital goods (machine) and end-of-life (EoL) of machine materials,
 - building infrastructure.

- **Excluded:**
 - fibre and yarn production (handled upstream),
 - textile/material waste (yarn losses and waste-yarn treatment), which is handled externally by the caller via the process-specific IOR,
 - building construction costs beyond the amortised area share modelled in Section 3.5,
 - packaging and distribution.
- **Background data:** ecoinvent 3.12, system model: cut-off by classification. Background processes are linked via generic market activities (GLO/RoW).

Analysis

2.1 Key findings from literature

1. Flat-knitting-specific data are extremely scarce.

Most studies either combine flat and circular knitting or report factory-level electricity averages without distinction between technologies.

2. Electricity values with the strongest evidence:

- **EcoBalyse:** 1.2–2.4 kWh/kg for knitting.
- **van der Velden et al. (2013):** ~1.16 kWh/kg for flat knitted fabrics.

3. Yarn losses (waste yarn during knitting):

- Typically **1 - 5%**, depending on yarn, machine efficiency and operator adjustments.
- EcoBalyse: **3 - 5.5%** (flat and circular knitting).

Sources: Laursen 2007; Sandin 2019; Roos 2015.

4. Lubricant consumption:

- Typical: **~0.005 kg/kg fabric.**
- Worst case: **0.008 kg/kg.**

Sources: Idemat 2012; Hasanbeigi & Price 2015; Sandin 2019.

5. Textile waste (trimmings, machine adjustments, errors):

- Around **1.5 - 5%**.
- EcoBalyse: **3 - 5.5%**.

Sources: Laursen 2007; Sandin 2019; Roos 2015.

6. Capital goods (machine) and EoL:

- EF and ecoinvent require infrastructure/capital goods to be included unless excluded with justification.
- Mass-based amortisation over lifetime output is the recommended approach.

7. Indirect energy (HVAC, compressors, lighting, cleaning):

- Sub-metered data rarely exist.
- Evidence from spinning and weaving shows that compressors, HVAC and auxiliaries may represent a substantial share of electricity consumption.

Process Inventory - Flat Knitting

3.1 Average Case – Flat Knitting (mix)

Functional unit: 1 kg knitted fabric

Category	Flow	Amount	Unit	Comment	Dataset
	Lubricant, average	0.005	kg	Typical oil consumption per kg (Idemat 2012; Hasanbeigi & Price 2015).	lubricating oil production {RoW} cut-off
	Air emissions from lubricant	0.005	kg	Mass-based approximation.	market for waste mineral oil {RoW} cut-off
Electricity / heat	Electricity mix	1.2	kWh	EcoBalyse mid-range (1.2–2.4 kWh/kg).	market group for electricity, medium voltage {GLO} cut-off
	Indirect Energy (Compressors , HVAC, Illumination, etc..)	4.46	kWh		market group for electricity, medium voltage {GLO} cut-off
Capital goods	Capital good (Building Machine)	2.8E-06	unit	Mass of machine allocated per kg of fabric (see section 3.4).	Refers to the ecoinvent 3.12 dataset market for building machine {GLO} cut-off , which

Category	Flow	Amount	Unit	Comment	Dataset
					represents generic industrial machinery. This is not a custom internal dataset, it is the standard ecoinvent proxy for process equipment amortisation.
Building infrastructure	Building infrastructure	0,000139	m^2		market for building, hall {GLO} cut-off
Waste to treatment	EoL Machine (Steel)	0.0047	kg	Steel fraction amortised per kg.	market for waste steel {RoW} cut-off
	Recycling others (Other Materials)	0.00083	kg	Remaining materials amortised per kg.	treatment of waste electric and electronic equipment, shredding {GLO} cut-off

3.2 Worst Case – Flat Knitting (mix)

Category	Flow	Amount	Unit	Comment	Dataset
Product	Coarse knitted tricot	1	kg		
	Lubricant	0.008	kg	Based on Sandin 2019 (p. 46).	lubricating oil production {RoW} cut-off
	Air emissions from lubricant	0.008	kg		market for waste mineral oil {RoW} cut-off
Electricity	Electricity mix	2.4	kWh	Upper EcoBalyse estimate; primary data	market group for electricity, medium

Category	Flow	Amount	Unit	Comment	Dataset
				preferred when available.	voltage {GLO} cut-off
	Indirect Energy (Compressors , HVAC, Illumination, etc..)	4.46	kWh		market group for electricity, medium voltage {GLO} cut-off
Capital goods	Capital good (Building Machine)	2.8E-06	unit	Same machine allocation until better factory data.	Refers to the ecoinvent 3.12 dataset market for building machine {GLO} cut-off , which represents generic industrial machinery. This is not a custom internal dataset, it is the standard ecoinvent proxy for process equipment amortisation.
Building infrastructure	Building infrastructure	0,000139	m^2		market for building, hall {GLO} cut-off
Waste	Steel EoL	0.0047	kg		market for waste steel {RoW} cut-off
	Other materials EoL	0.00083	kg		treatment of waste electric and electronic equipment, shredding {GLO} cut-off

3.3 Best Case – Flat Knitting (mix)

Category	Flow	Amount	Unit	Comment	Dataset
Product	Coarse knitted tricot	1	kg		
	Lubricant	0.005	kg	Standard default when primary data unavailable (Idemat 2012).	lubricating oil production {RoW} cut-off
	Air emissions from lubricant	0.005	kg		market for waste mineral oil {RoW} cut-off
Electricity	Electricity mix	1.16	kWh	van der Velden et al. 2013 (efficient case).	market group for electricity, medium voltage {GLO} cut-off
	Indirect Energy (Compressors , HVAC, Illumination, etc..)	4.46	kWh		market group for electricity, medium voltage {GLO} cut-off
Capital goods	Capital good (Building Machine)	2.8E-06	unit		Refers to the ecoinvent 3.12 dataset market for building machine {GLO} cut-off , which represents generic industrial machinery. This is not a custom internal dataset, it is the standard ecoinvent proxy for process equipment amortisation.
Building infrastructure	Building infrastructure	0,000139	m^2		market for building, hall

Category	Flow	Amount	Unit	Comment	Dataset
					{GLO} cut-off
Waste	Steel EoL	0.0047	kg		market for waste steel {RoW} cut-off
	Other materials EoL	0.00083	kg		treatment of waste electric and electronic equipment, shredding {GLO} cut-off

3.4 Capital Goods (Machine)

Shared methodology: The full modelling framework, calculation logic, and default values for capital goods and end-of-life are documented in [Capital Goods & End-of-Life \(Machinery\) in LCA](#). This section states only the flat-knitting-specific inputs.

We allocate a very small fraction of the machine's mass to each kilogram of fabric, consistent with EF rules and ecoinvent practice.

3.4.2 How it works

1. Take the **machine mass** (e.g., 1000 kg).
2. Estimate total lifetime fabric output (years × hours/day × days/year × kg/h).
3. Divide machine mass by lifetime output:

The formula for amortising the machine mass per kg of fabric is:

$$\text{kg machine per kg fabric} = \frac{M_{\text{machine}}}{Q_{\text{lifetime}}}$$

Where:

- **M_{machine}** = Total mass of the machine (kg)
- **Q_{lifetime}** = Total lifetime production output (kg fabric)

The lifetime output is calculated as:

$$Q_{\text{lifetime}} = \text{Years} \times \frac{\text{Hours}}{\text{Day}} \times \frac{\text{Days}}{\text{Year}} \times \frac{\text{kg fabric}}{\text{Hour}}$$

For the average scenario:

- 15 years
- 16 hours/day
- 300 days/year
- **2.5 kg/h productivity** (verified: Sino Finetex industry benchmark for computerised flat knitting: 10–20 kg/day; Changhua Knitting Machine: 30 min–8 h per sweater at 300–500 g each → mid-range ~2.5 kg/h. Sources: sinofinetex.com; changhua-knitting-machine.com)
- **Machine mass: 1000 kg** (verified: Fengfan double-carriage computerised flat knitting machine, 2500×850×1900 mm, net weight ~1000 kg. Source: indiamart.com listing; Fibre2Fashion supplier: 2000×840×1870 mm → 1000 kg)

This yields:

0.0056 kg machine / kg fabric

Calculation: $1000 \text{ kg} \div (15 \text{ yr} \times 16 \text{ h/d} \times 300 \text{ d/yr} \times 2.5 \text{ kg/h}) = 1000 \div 180,000 = \mathbf{0.0056 \text{ kg/kg}}$

Source: Machine dimensions (2700 mm × 909 mm) confirmed from KARL MAYER STOLL CMS 503 ki L official brochure (stoll.com PDF). Machine mass (1000 kg) based on verified OEM datasheets for comparable computerised flat knitting machines (Fengfan double-carriage, Fibre2Fashion listings) — STOLL does not publish machine weight in public brochures. Productivity (2.5 kg/h) consistent with industry benchmarks for computerised flat knitting (Sino Finetex 2025; Changhua Knitting Machine). Operational parameters (15 years, 16 h/day, 300 days/year) are mid-range engineering defaults bracketed by WCS/BCS scenarios; consistent with Hasanbeigi (2010) and IPPC BREF (2003). Replace with primary factory data when available.

3.4.3 End-of-life of the machine

Machine material split:

- **85% steel** → 850 kg
- **15% other materials** → 150 kg

Amortised as ($\div 180,000 \text{ kg}$ lifetime output):

- **Steel EoL = 0.0047 kg/kg fabric**
- **Other materials EoL = 0.00083 kg/kg fabric**

3.4.4 How this goes into Activity Browser

The dataset "**market for building machine**" uses **unit**, not kg.

- 1 unit = one generic entire machine (2000 kg)

- We convert our kg/kg figure into a *fraction of a unit*:

$$X = \frac{0.0056}{2000} = 2.8 \times 10^{-6} \text{ unit per kg fabric}$$

In ecoinvent (version 3.12):

- **Amount** = 2.8E-06
- **Unit** = unit
- **Product** = building machine

3.5 Building Infrastructure

Shared methodology: The full framework is documented in [Building Infrastructure in LCA](#). This section states only the flat-knitting-specific inputs.

Process-specific value: 1.39E-04 m²/kg — derived from KARL MAYER STOLL CMS 503 ki L envelope (2700 mm × 909 mm) with cell factor accounting for aisles, yarn storage and maintenance access. Flat knitting has markedly lower throughput per allocated space than circular or seamless due to panel-by-panel operation with stop-and-start cycles.

ecoinvent activity: market for building, hall {GL0} | cut-off (unit: m²)

3.6 Indirect Energy (HVAC, Compressors, Lighting, Cleaning)

Shared methodology: The full derivation, evidence base, and subsystem distribution are documented in [Indirect Energy in LCA — Shared Methodology \(Knitting\)](#).

Default value: 4.46 kWh/kg — derived from the upper EcoBalyse estimate for knitting machine electricity (2.4 kWh/kg = 35% of total plant electricity; ADEME & Ministère de la Transition Écologique, 2024): $2.4 \div 0.35 \times 0.65 = 4.46 \text{ kWh/kg}$.

This value is **identical for flat and circular knitting** and is a temporary reference pending sub-metered primary data. See the [shared page](#) for subsystem distribution and full evidence base.

Input–Output Ratio (IOR)

Definition: $\text{IOR} = 1 / (1 - L)$, where L is the yarn loss rate (as a fraction). For every 1 kg of finished knitted fabric, the upstream yarn input must be scaled by IOR.

How the loss rate is derived

Flat knitting operates panel-by-panel with stop-and-start cycles, garment shaping, and yarn carrier changes. Yarn losses arise from thread breaks, trial runs, edge waste, and machine setup. The 1.5–5% range brackets best-case (modern machines, simple panels) to worst-case (fine gauge, complex shaping). The 3% average is the default used in the LCI dataset. Sources: Laursen et al. (2007); Sandin et al. (2019); EcoBalyse (ADEME & Ministère de la Transition Écologique, 2024) — range 3–5.5%.

IOR values

Scenario	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per kg output (kg)
BCS (Best Case)	1.5%	1.0152	1.0152	0.0152
Average (default)	3%	1.0309	1.0309	0.0309
WCS (Worst Case)	5%	1.0526	1.0526	0.0526

How to apply: for 1 kg of flat knitted fabric, multiply upstream yarn input by the IOR. Example at average: 1 kg fabric requires **1.031 kg yarn input**; 0.031 kg is lost as waste yarn and routed by the caller to for example `market for waste yarn and waste textile {RoW}`. No inline yarn-loss or textile-waste exchange is included in the dataset itself.

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